Week 4 Part E — Written justification

Scope and integrity. The figure uses the complete recorded daily series (1 April 2024–31 August 2025 for the trend panel); the partial September 2025 tail is excluded. Part B tested 60 start/window combinations: **34 slopes rose and 26 fell**, so trend direction is sensitive to framing. I use the 7-day window because the diagnosed recurring cycle is weekly, not because it maximises a preferred trend. The start date is the first recorded day; no later start was selected. Part B reports the 28-day alternative (35.7% variation removed versus 32.4% for 7 days); it smooths more but also blurs month-scale movement. This sensitivity should accompany, not be hidden behind, the selected view.

Four panels and Week 2 channel rules. (1) The daily line asks whether traffic level changes after weekly variation is damped; a line chart fits ordered time, with position on a common axis and a distinct line for raw versus smoothed values. (2) The weekday × hour heatmap asks when traffic is relatively high or low; a matrix with a sequential light-to-dark colour channel encodes ordered magnitude, while weekday and hour remain labelled. (3) The treemap asks how net revenue is distributed through the category hierarchy; nested area represents part-to-whole magnitude and hue distinguishes parent categories. (4) Sorted horizontal bars ask how subcategories compare and rank; common-baseline position/length is more accurately compared than area or angle, and direct numeric labels make close values checkable.

Time decisions. Timestamps were parsed by explicit format: ISO strings as `%Y-%m-%d %H:%M:%S`, slash dates as day-first `%d/%m/%Y %H:%M`; 0 parse failures. This avoids ambiguous inference that could shift dates. After parsing, rows were sorted and duplicate timestamps deduplicated. Hourly sessions are a **flow** (events counted during each hour), so daily values are the **sum** of valid hourly counts, not the last hourly value. The 7-day rolling mean is **centred** because this is retrospective review and the weekly cycle is the target; centring balances observations before and after each date. It is not suitable as a live forecast because it uses future observations. Raw daily totals remain visible. The source covers 1 April 2024–11 September 2025, but September is incomplete (11 calendar days), so it is excluded from trend comparisons rather than treated as a full month.

Rows and missingness. Session file: 12,670 input rows → 12,384 unique timestamps after removing **286 duplicate logger rows** (identical timestamp/metrics/device; keep one, do not double-count). Of the deduplicated rows, **287 `sessions = -1` sentinel values** were changed to missing, not deleted; they are excluded from sums. No timestamp parse failures. The timestamp span contains 12,696 expected hourly slots; 12,384 were recorded, leaving **312 absent hours** (not fabricated as zero). Daily aggregation retains **13 dates with no valid hourly sessions as missing (`NaN`)**. The incomplete September tail comprises 11 daily slots excluded from the trend panel. Catalogue: 345 rows → 344 after removing **1 duplicate SKU** (keep first; repeated item/revenue would otherwise be counted twice). **Seven negative net-revenue rows are retained**, because returns can exceed sales; they are not “bad rows.”

Hierarchy and roll-up. No subcategories were rolled into “Other”: the catalogue has **4 parent categories and 8 subcategories**, and the chart keeps the hierarchy to its available second level (category → subcategory). Therefore `Other` contains **0 categories and PKR 0m**. The eight subcategory net-revenue totals sum to **PKR 1,711.382m** after SKU deduplication; the treemap's area layout clips negative values to zero because negative area is not meaningful, while the sorted bar panel shows the signed values. No deeper SKU-level roll-up is shown. After SKU deduplication, the eight subcategory totals sum to PKR 1,711.382m. The raw, pre-deduplication total is PKR 1,726.209m; it is not the chart denominator.